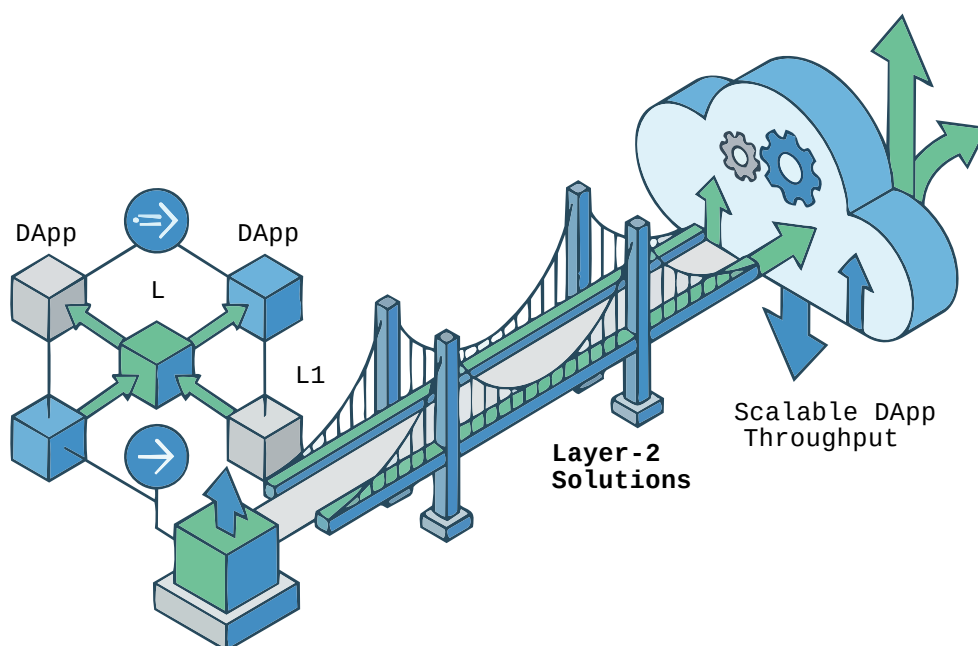


Designing Scalable dApps with Layer-2 Solutions

Layer-2 solutions are changing how scalable decentralised apps are made by fixing the problems with Layer-1 networks like Ethereum.



Decentralised, trustless systems that don't need middlemen have been made possible by Web3 applications (dApps) built on blockchain platforms like Ethereum. But as more people use these networks, they will face a big problem: scalability. Scalability is the ability of a blockchain to handle a lot of transactions quickly and cheaply without losing speed or quality.

Every transaction on Layer-1 blockchains like Ethereum must be checked and processed by all the nodes that are part of the network. This makes the system very secure and decentralised, but it also makes it very slow. Ethereum can only handle about 15 to 30 transactions per second (TPS), which is much lower than traditional systems like Visa, which can handle thousands of TPS. Because of this, users have to pay high gas fees and wait a long time for transactions to be confirmed when demand on the network goes up.

What does Layer-2 mean (without the buzzwords)?

Layer-2 is like a helper system that sits on top of a blockchain like Ethereum. It speeds things up and makes them cheaper without changing the core network itself. In a dApp, every transaction, like sending tokens or interacting with a smart contract, must be done directly on Ethereum (Layer-1). This method is safe, but it takes a long time and costs a lot of money because thousands of nodes have to check each transaction. Layer-2 takes care of most of the work outside of the main blockchain and only sends the final result back to it.

A simple way to think about it is to picture a busy highway (Layer-1) where all the cars have to go down the same road, which causes traffic jams. Layer-2 is like a service road or flyover, where most cars go their own way and only connect to the main highway when they need to.

Types of Layer-2 solutions

There are different types of Layer-2 solutions, and each one is made to make Layer-1 blockchains like Ethereum more scalable while keeping them safe. Optimistic rollups, ZK rollups, and state channels and sidechains are the most common types.

Optimistic rollups: Optimistic rollups assume that transactions are valid by default ('optimistically') and only check them if someone questions the outcome. Instead of processing each transaction on its own, they group several transactions together and send them to Ethereum as one batch.

ZK rollups: Zero-Knowledge (ZK) rollups use cryptographic proofs to check transactions in a different way. They don't just assume that transactions are valid; they make a mathematical proof (called a validity proof) that guarantees correctness before sending it to Layer-1. ZK rollups are used by solutions like zkSync and StarkNet.